

Project Log: Design Decision

Project: Smart Campus Network Optimization Platform

Log Date: September 20, 2026

Log Type: Decision / Architecture Design Review

What Changed

The AI automation and monitoring n8n workflow (Team 2's module) was reviewed node by node to decide which of the originally placeholder "agent" blocks genuinely need a real LLM and which should be plain deterministic nodes. Several nodes were reclassified and the following adjustments were made to the design:

- **Kept as true LLM agents:** Command Sorting System (routes free-text chat commands) and the Configuration Agent (turns natural language into executable device configuration).
- **Reclassified as deterministic, non-LLM nodes:** Day 0 Protocol, Credentials Machine, Network Write, and the AWS Cloud upload step — these are now scoped as scripts/database lookups, not AI agents.
- **Code Validator adjusted:** primary check is now a deterministic linter/dry-run against the target vendor's syntax; an LLM review is kept only as an optional secondary opinion, not the sole gate before a config is pushed to a production switch.
- **Decision Maker adjusted:** branch selection (chat log request vs. timer trigger vs. health segment vs. attack segment) moves to plain n8n IF/Switch logic; a light LLM call is kept only for phrasing the final chat-facing reply.
- **Network Health Agent adjusted:** default behavior changed to a templated alert message ("Port X on device Y went down at Z"); an LLM is invoked only when more varied or explanatory phrasing is needed.
- **Attack Analysis Agent adjusted:** a dedicated trained ML classifier (Random Forest / XGBoost / Isolation Forest, trained on a labeled dataset such as CICIDS2017, NSL-KDD, or UNSW-NB15) is added upstream of the LLM. The classifier performs the actual attack detection/classification; the LLM's role is narrowed to explaining the classifier's verdict in the chat alert.
- **Free-tier LLM API providers selected** for each remaining agent role, after surveying 15 candidate providers: Groq (Llama-3.3-70B-Versatile) for the Automation system (Command Sorting + Configuration Agent); Google AI Studio (Gemini 3.1 Flash) for the Decision Maker's reply phrasing and Data Logs summarization; OpenRouter (routed to free DeepSeek R1 / Llama 3.3 70B) for the Attack Analysis explanation layer.

Resulting Design: Revised Architecture Diagrams

The two diagrams below reflect every adjustment listed above — deterministic nodes separated from true LLM agents, and the attack-detection path backed by a trained classifier instead of a raw LLM guess.

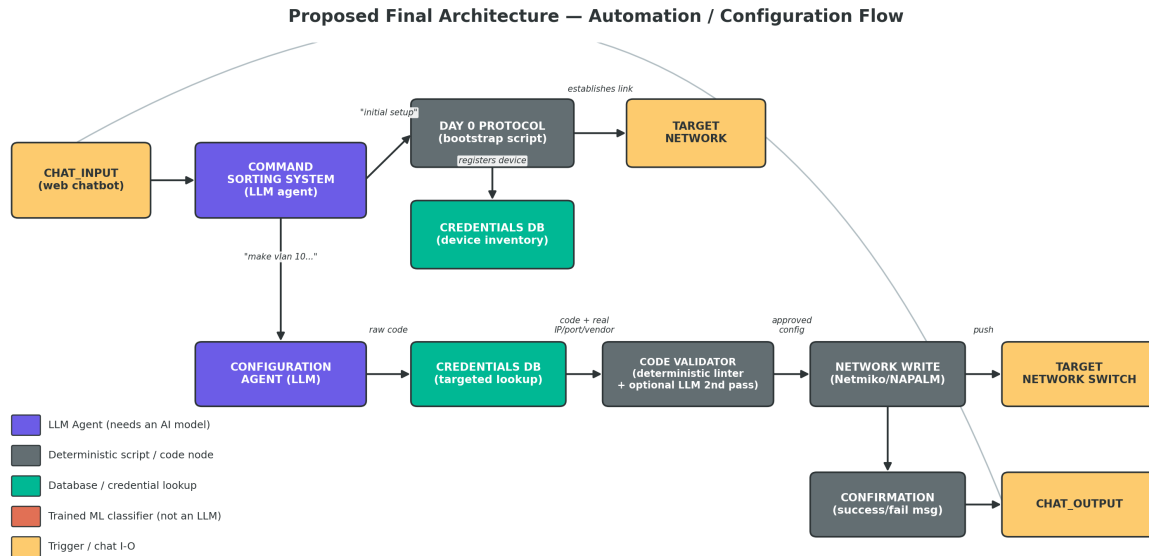


Figure 1 — Revised architecture: Automation / Configuration flow.

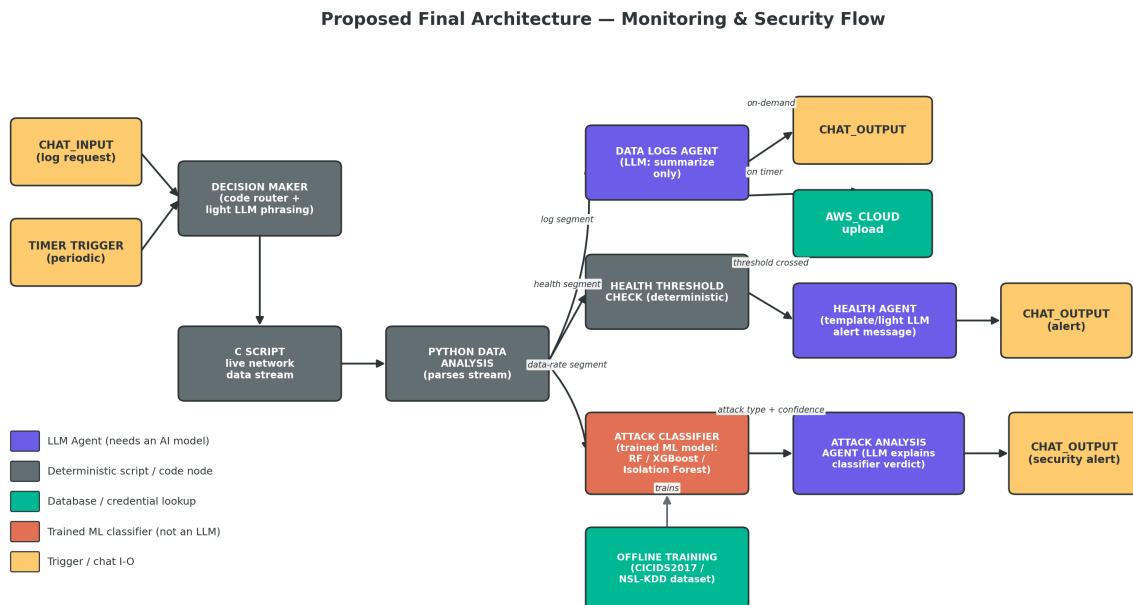


Figure 2 — Revised architecture: Monitoring & Security flow.

Reasoning / Notes

The guiding test applied to every node: an LLM adds real value where a node has to turn language into structure or structure into language, and adds unnecessary risk, latency, or cost where a node is a deterministic action, a credential/database lookup, or has to interpret raw numeric telemetry.

Network configuration correctness and device credential handling were treated as too safety-critical to leave to language-model judgment alone, so both are now backed by deterministic checks and lookups rather than an agent's best guess. Attack classification was specifically flagged as unreliable if left to an LLM reading raw data-rate numbers — it can produce confident-sounding but wrong classifications — so a supervised classifier was added as the actual detection mechanism, which also lines up with the simulated-attack training dataset option already floated for this module.

Provider selection prioritized reliability across the full project build (Sep 2026 – Jun 2027) over raw benchmark performance: a permanent no-card free tier for the two highest-traffic roles (automation, decision routing), and provider redundancy (via an aggregator) specifically for the attack-analysis path, since that is the one place a single provider going down or rate-limiting would be most costly.

Impact on Project Plan

- Team 2's deliverables now include a small offline-trained ML classifier for attack detection, in addition to the LLM-based chatbot pipeline already scoped in the September 12, 2026 initiation log.
- The Jan 15, 2027 "Pi Becomes Central Host" and Feb 15, 2027 "Mid-Project Review" milestones should account for this addition when Team 2 plans classifier training-data sourcing and integration.
- No change to Team 1 (Wi-Fi optimization) or Team 3 (network topology) scope; this log affects only Team 2's automation/monitoring/AI module. Team 3's Cisco-only network scope from the September 12, 2026 decision log is unchanged and unaffected by this design review.
- Supporting artifacts produced alongside this decision: a full workflow summary report (original design, revised architecture diagrams, node-by-node review) and a comparison spreadsheet of 15 free LLM API providers.

Status

Design review complete for the AI automation and monitoring workflow. Revised architecture diagrams and provider selections recorded. Classifier training-data sourcing (CICIDS2017 / NSL-KDD / UNSW-NB15) and integration testing against the Cisco-only network (per the September 12, 2026 decision log) to be confirmed in a future log once implementation begins.